ELE 637 - Pre-Lab # 2

Stoi Mourad
501170415

(1)

Additive: Magnetic flux created by the current in one winding in the same as the flux created by the current in the other winding.

Subtractive: Fluxes created by currents in the coupled windings flow in opposite directions

(2) Output of electric machines' mechanical output is rated in kW (kilowatts)/ horse power → represents active/real power

Transformers are rated in KVA (both active + reactive power). KVA is used for proper sizing the windings/cables of the transformer.

(3) Rated Voltage: Max voltage that can be applied to the transformer without causing excessive heating in the windings.

Rated current: Max current that can be applied to a transformer without causing excessive heating in the windings.

(4)

The open-circuit test is preformed on LV side since:

(1) Test is to measure transformer core that requires applying a full voltage (rated voltage). This minimizes error in measurement.

(2) Testing equipment source usually has a limited voltage that may not be as high as the transformer

The short-circuit test is preformed on the HV side since:

(1) Test is to measure transformer winding that requires a full current (rated current). This minimizes error in measurement.

(2) Testing equipment source usually has a limited current that may not be as high as the transformer.

(3) less stressful test in lower current.

(5) $R_L = 20\Omega$, $R_L' = 20\left(\frac{600}{400}\right)^2 = 45\Omega$

(6) non-linear since as the core flux increases, the core can be saturated.

(7) This core can be used for electrical isolation of that input + output terminals.

(8) Eddy current loss + hysteresis loss can be lumped together P_{core}

Power per unit volume; $\frac{P_{core}}{V_{core}} = f \oint H\,dB$

$v = e = N\frac{dB}{dt} \rightarrow V \propto B$

∴ when the frequency + applied voltage are doubled, P_{core} will be four times higher. with only a double in voltage, P_{core} will double (before saturation),

(9) Y/Y, Y/Δ, Δ/Y, Δ/Δ